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Original Research

Does a recent hamstring muscle injury affect the timing of muscle activation during high speed overground running in professional Australian Football players?

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ABSTRACT

Objectives: To investigate if the temporal characteristics of hamstring and gluteal muscle activation are altered during high speed overground running in professional Australian Football players following hamstring muscle injury.
Design: Cohort study.
Setting: Field-based testing.
Participants: Elite professional Australian Football players who had sustained a hamstring muscle injury in the six months prior to testing (n = 7) and a group of players from the same club who had no history of hamstring muscle injury (n = 8).
Main outcome measures: Muscle onset timing, muscle offset timing and muscle onset duration of the medial hamstrings, biceps femoris and gluteus maximus muscles during high-speed running using electromyographic data.
Results: No significant differences in any of the temporal aspects of muscle activation were found between groups for any of the muscles tested ($p > 0.05$).
Conclusions: Persistent alterations to the timing of muscle activation following hamstring muscle injury that have been reported in recreational athletes were not observed during high speed running in professional athletes who have completed comprehensive rehabilitation programs.
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1. Introduction

Hamstring muscle injuries are the most prevalent injury in professional men's football (Ekstrand, Waldén, & Hägglund, 2016). These injuries lead to significant player unavailability (Woods et al., 2004), decreased performance levels (Hägglund et al., 2013) and place a significant financial burden on clubs (Hickey, Shield, Williams, & Opar, 2014). Despite ongoing efforts at improving rehabilitation and prevention protocols there remains a high rate of injury recurrence (van Heumen et al., 2017).

The risk of injury recurrence is influenced by factors such as how recently and the number of times a previous hamstring injury was sustained by an individual (Green, Bourne, van Dyk, & Pizzari, 2020). Hamstring muscle injuries can lead to structural changes within the muscle such as local atrophy (Sanfilippo, Silder, Sherry, Tuite, & Heiderscheit, 2013) and scar tissue formation (Silder, Heiderscheit, Thelen, Enright, & Tuite, 2008). These factors may contribute to persistent physical strength deficits (Croisier, Forthomme, Namurois, Vanderthommen, & Crielaard, 2002), reduced tissue extensibility (De Vos et al., 2014), and shortened muscle fascicles (Timmins et al., 2016), which have been proposed to influence hamstring injury recurrence.

Neuromuscular inhibition is one sequelae to hamstring muscle injury that has been proposed to contribute to an increased risk of re-injury (Fyfe, Opar, Morgan, & Shield, 2013). Following pain or injury, neuromuscular inhibition in the form of aberrant muscle timing may develop (Clark & Burden, 2005; Falla, 2004; Tucker & Hodges, 2009) resulting in changes to the control and function of the affected area (Hodges & Tucker, 2011). Changes to the temporal characteristics of muscle activity have been found to exist following

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injury, even after symptoms are resolved (Crow, Pizzari, & Buttifant, 2011). It has been proposed that following hamstring injury, neural inhibitory mechanisms are most evident during maximal eccentric contractions (Fyfe et al., 2013), which corresponds to the activity of the hamstring and gluteal muscles during the late swing phase of gait when the hamstring is most vulnerable to injury (Chumanov, Schache, Heiderscheit, & Thelen, 2012).

Evidence for the value of assessing neuromuscular activation for predicting index strains is variable. It has been reported that healthy athletes with delayed onset of their hamstring muscle in a prone hip extension exercise (Schuermans, van Tiggelen, & Witvrouw, 2017a), or who have lower amounts of gluteal muscle activity during overground maximal acceleration (Schuermans, Danneels, van Tiggelen, Palmans, & Witvrouw, 2017b), were more likely to suffer a hamstring strain in the following 18 months. In contrast, another study found that onset timing of the biceps femoris and medial hamstrings during isokinetic strength testing in healthy athletes was unrelated to risk of hamstring muscle injury in the following season (van Dyk et al., 2018). Another study reported that professional Australian Football players were more likely to sustain a hamstring muscle injury during the season if they had higher gluteus medius muscle activity during treadmill running prior to the season (Franettovich Smith et al., 2017). The same study found no association between activity levels of gluteus maximus during treadmill running and future hamstring muscle injury.

There is mixed evidence that the muscle activation of the hamstring and gluteal muscles may be altered following injury. One recent study of collegiate sprinters with a history of hamstring muscle injury found reduced biceps femoris activation during late swing phase in the previously injured side compared to the uninjured side during maximal overground acceleration, but not during other phases of the running cycle, or in the gluteus maximus (Higashihara et al., 2019). Altered ratios of biceps femoris to gluteus maximus EMG activity have been reported as being isolated to the late swing phase of fast treadmill running in athletes with a history of hamstring muscle injury (Daly, McCarthy Persson, Twycross-Lewis, Woledge, & Morrissey, 2016). However, another study found no changes in the activation patterns of biceps femoris and medial hamstrings during fast treadmill running in athletes with a history of hamstring muscle injury (Silder, Thelen, & Heiderscheit, 2010).

Altered muscle activation in tasks other than running have also been investigated. Earlier onset timing of the biceps femoris and medial hamstrings has been reported during transition from double leg to single leg stance in males with a history of hamstring muscle injury (Sole, Milosavljevic, Nicholson, & Sullivan, 2012) and a lower rate of torque development of the biceps femoris has been reported during isokinetic testing in recreational athletes with a history of hamstring muscle injury (Opar, Williams, Timmins, Dear, & Shield, 2013).

The specific effects of hamstring muscle injury on the temporal characteristics of the hamstring and gluteal muscles during fast overground running remains unclear. Although previous studies have isolated changes in muscle activation levels to certain parts of the running cycle (Daly et al., 2016; Higashihara et al., 2019) there have been no studies to date that consider the specific temporal characteristics of hamstring and gluteal muscles activation during fast overground running.

The objective of this study was to investigate whether the muscle onset time, muscle offset time or duration of muscle activation in the biceps femoris, medial hamstrings or gluteus maximus are altered in high speed running following hamstring muscle injury in professional Australian Football players compared to those with no hamstring muscle injury history. The study also aimed to provide a graphical representation of muscular activity throughout the gait cycle to assist in the interpretation and understanding of the results.

2. Materials and methods

2.1. Participants

Professional male Australian Football players from a single AFL club were recruited. In total, 15 players were included in the study; seven players with a history of hamstring muscle injury within the preceding six months (HS group) and eight control participants (CON group). All previous hamstring injuries were diagnosed by medical staff at the AFL club and confirmed by magnetic resonance imaging (MRI). The most recent strain injury occurred seven weeks prior to testing. All control participants had no recorded history of a hamstring injury while contracted to play in the AFL and reported having never sustained a hamstring injury prior to the study. All participants were in full pre-season training at the time of the study, reported having no history of lower limb surgery, and had no recorded lower limb surgery while contracted to play in the AFL.

The study protocol was approved by the Faculty Human Ethics Committee (FHEC11/170), and all participants provided written informed consent.

2.2. Injury questionnaire

Participants completed a short questionnaire, which included demographic information as well as information about their injury history. Collected data were cross-referenced with comprehensive injury data maintained by the club to confirm the accuracy of self-reported injury histories.

2.3. Instrumentation and electrode placement

Surface EMG electrodes consisted of Trigno (Delsys Inc., Boston, USA) wireless sensors with a single differential configuration and a 4 bar (99.9% silver) contact area, with an inter-electrode distance of 10 mm.

Skin preparation and electrode placement were consistent with recommended SENIAM guidelines (Hermens, Freriks, Disselhorst-Klug, & Rau, 2000). Gluteus maximus electrode placement was midway along a line between the greater trochanter and the sacral vertebrae. The biceps femoris electrode placement was midway along a line between the ischial tuberosity and the lateral epicondyle of the tibia, and medial hamstring electrode placement was midway along a line between the ischial tuberosity and the medial epicondyle of the tibia.

Footswitches (Model: 402, Interlink Electronics, USA) were positioned bilaterally on the plantar aspect of the heel and interphalangeal joint of the hallux, and used to record temporal aspects of the gait cycle (Semciw, Pizzari, Murley, & Green, 2013).

A Delsys Trigno Wireless 16-Channel EMG system was used to collect raw EMG signals (Delsys Inc., Boston, USA; CMRR > 80 dB @60 Hz; gain of 1000; band pass filtered at 20–450 Hz) and sampled at 2000 Hz.

Players wore a 5Hz Global Positioning System (GPS) unit (MinimaxX, Team 2.5, Catapult Innovations, Australia) in a specifically tailored garment during the testing protocol to assess maximum running speed during each trial (Jennings, Cormack, Coutts, Boyd, & Aughey, 2010).

Ambient temperature was recorded using a portable weather metre (Kestrel 2500; Kestrel Instruments, PA, USA) during the warm-up that preceded each trial.

2.4. Study procedure

The testing procedure was conducted on a 100m stretch of grass within an outdoor field at La Trobe University, Melbourne, Australia. To negate any possible effects of fatigue or muscle soreness, participants did not engage in any intensive training within the 48 h prior to testing. Following EMG electrodes, footswitch and GPS application, the participant completed a standardised warm-up protocol consisting of 200 m of jogging followed by 100 m walk recovery. The participants next completed 100 m runs at 60%, 70%, and 80% of their perceived maximum speed consecutively with 100 m walk recovery between each effort. Participants then completed one practice trial of the testing procedure before commencing testing.

The running trial protocol was adapted from a graded running protocol developed for hamstring rehabilitation (Reid, 1993). Participant running trials were divided into a 40 m acceleration section where players commenced from a standing start and were instructed to steadily build up their speed to 90% of their perceived maximum, a 20 m middle-section where players were instructed to hold their speed at 90% of their perceived maximum, and a 40 m deceleration section where players were instructed to steadily reduce their speed so that they were walking by the end of the run. Participants then walked back the full 100 m before commencing their next trial.

Two sessions of testing were undertaken within a seven-day period and participants were randomly assigned to one of the two testing sessions.

2.5. EMG data processing and analysis

All EMG data were processed using a customised adaptation of the Biosignal EMG package (2.1.0) from R statistical software (https://www.r-project.org/, version 3.6.1). Raw EMG signals for each running trial were high pass filtered (Butterworth, 4th order, zero phase lag, 20Hz cut-off) to remove movement artefact. Data were full wave rectified and further filtered using a moving average (20 ms window). Two consecutive strides from the middle 20m of each running trial were further processed for analysis (two strides x five trials = ten strides per participant). These particular strides were chosen to decrease the likelihood that participants were accelerating or decelerating during the stride, and increase the likelihood that participants were at their highest speed at the point of analysis. In the HS group the most recently injured side was chosen for analysis. In the CON group the players preferred kicking leg (skill leg) was chosen for analysis. Individual strides were considered to start at heel contact, detected using the heel footswitch, and end at the next consecutive heel contact on the same side. The data from the gait cycle were time-normalised to 101 points (representing the gait cycle from 0 to 100% in 1% increments) enabling time-dependant analysis between groups (Schuermans, Danneels, van Tiggelen, Palmans, & Witvrouw, 2017a). Data were amplitude normalised to peak activity recorded through the running cycle.

For each muscle and participant, an ensemble average was generated from the ten strides. All participants' ensemble averages were summed and averaged to produce a grand ensemble for gluteus maximus, biceps femoris and medial hamstrings in both the HS group and the CON group. This allowed an EMG profile to be established for each muscle in each group across the high-speed running gait cycle. For each muscle and participant, the muscle onset time, muscle offset time and duration of muscle activation were calculated as a percentage of gait cycle. Muscle onset time was calculated using a customised algorithm in the R Statistical software package (version 3.6.1) and determined as the time when EMG amplitude was increased above baseline by a level of $\geq$15% of peak amplitude for a period of $\geq$10% of the gait cycle. The time of muscle offset was determined as the point when EMG amplitude returned to baseline or remained below 15% of the maximum EMG amplitude for $\geq$10% of the gait cycle (Chapman et al., 2006; Franettovich, Chapman, & Vicenzino, 2008); . Duration of muscle activation was the percentage of the gait cycle contained between the muscle onset and muscle offset.

Descriptive statistics (means, medians, interquartile range and standard deviations) were calculated and data reviewed for normality using boxplots and the Kolmogorov-Smirnoff (K–S) test. To compare each temporal variable for each muscle between groups, a Mann-Whitney *U* test was used, since data were not normally distributed. The comparison between demographic data was done by independent groups *t*-test.

To determine the magnitude of any differences between groups, a standardised effect size was calculated by dividing the z-score of the Mann–Whitney *U* test by the square root of the total sample size (Field, 2009). Effect sizes were categorised as small (0.2), medium (0.5) and large ($\geq$0.8) (Sullivan & Feinn, 2012). All statistical comparisons were performed using the SPSS statistical software package (Version 19, IBM SPSS Inc., Chicago, IL, USA).

3. Results

Participant demographic information, environmental conditions during testing, and running speeds, are displayed in Table 1. There was no statistical difference between HS and CON groups on any of these measures.

The footswitch data of one participant (HS group) was discovered to be corrupted after testing was completed and consequently that participant's gait was unable to be analysed. Consequently, after that participant's data was discarded, analysis was conducted on the gluteus maximus, medial hamstrings and biceps femoris of remaining six participants in the HS group. With the exception of one electrode on one participant's medial hamstrings (CON group), which dislodged during testing, electrodes remained in place during the testing sessions. The biceps femoris and gluteus maximus were analysed for all eight participants from the CON group, and the medial hamstrings were analysed for seven of the eight participants in the CON group.

3.1. Gait

Fig. 1 contrasts the grand ensemble curves between the HS group and the CON group for the gluteus maximus, biceps femoris and medial hamstrings.

3.2. Temporal characteristics

No statistically significant differences were observed between HS and CON groups for muscle onset, muscle offset or the duration of muscle activation for gluteus maximus, biceps femoris or medial hamstrings between the HS group and CON group (Table 2).

4. Discussion

This study did not find any differences in the temporal characteristics of muscle activation during high speed overground running between professional Australian Football players with a recent history of hamstring muscle injury and those with no prior history of hamstring injury.

The findings of this study support the findings of previous research that reported no differences in muscle onset time, muscle offset time or the duration of muscle activity between injured and

Table 1
Participant demographic and testing information.

Variable	HS group ($n = 7$) Mean (SD)	CON group ($n = 8$) Mean (SD)	p value
Age (years)	22.3 ± 1.6	21.9 ± 1.6	0.651
Height (centimetres)	189.1 ± 4.3	187.1 ± 3.4	0.362
Weight (kilograms)	90.6 ± 3.8	88.6 ± 2.8	0.306
Average running speed (m.sec^{-1})	7.3 ± 0.3	7.2 ± 0.4	0.848
Ambient temperature (°C) at testing	20.5 ± 2.2	23.4 ± 3.4	0.095
Time since HS strain (days)	108 ± 46	–	–

Abbreviations: SD, standard deviation; °C, degrees celsius.

uninjured limbs during fast treadmill running following hamstring muscle injury (Silder et al., 2010). Changes to the temporal characteristics of muscle activation have been reported to occur during isokinetic testing (Opar et al., 2013) and a double-leg to single-leg stance task (Sole et al., 2012) following hamstring muscle injury. Given that most hamstring muscle strains occur during high-speed running (Woods et al., 2004), and any neural inhibitory mechanisms are most likely to be present during strong eccentric contractions (Fyfe et al., 2013; Sole, Milosavljevic, Nicholson, & Sullivan, 2011), it is proposed that due to its specificity, high-speed running is the most appropriate task to assess motor patterning as a risk factor for future injury risk.

Other studies have reported altered levels of muscle activation in specific phases of the gait cycle during fast running (Daly et al., 2016; Higashihara et al., 2019). However, our study chose not to focus on muscle activation levels or use maximal voluntary isometric contractions in the analysis as we wanted to evaluate running patterns as opposed to strength or muscle activation levels. It is acknowledged that other parameters of muscle activation such as normalised amplitude or pattern of activity may have yielded different results. Timing of events in the gait cycle may be influenced by which threshold is used to determine onset and offset. This study used an onset and offset threshold of 15% of peak amplitude, an approach that has been used previously to assess changes in the temporal characteristics of muscle activity between groups (Chapman et al., 2006; Franettovich Smith, Honeywill, Wyndow, Crossley, & Creaby, 2014).

All participants in this trial were professional athletes who received a comprehensive hamstring rehabilitation program following their hamstring muscle injury. It is possible that any changes to the temporal characteristics of hamstring or gluteal muscle activation that may have been present following injury ceased either when pain related to the injury resolved or as a result of a comprehensive rehabilitation process. Both previous studies identifying residual adaptations in muscle onset time following hamstring muscle injury were in recreational athletes that may not have completed a comprehensive rehabilitation program (Opar et al., 2013; Sole et al., 2012).

This study was conducted in an environment designed to closely simulate that of a professional football match. Differences in muscle activation have been observed between treadmill and overground gait (Anders, Patenge, Sander, Layher, & Kinne, 2019) and consequently overground gait was investigated in this trial. The high-speed running was conducted outdoors on a grass field to replicate the surface encountered by players on matchday. However, as the trial was conducted over two testing sessions, for practical reasons, environmental conditions such as temperature, wind and ground hardness were not able to be controlled as well as they might have been in a laboratory setting. For example, the CON group ran in slightly hotter conditions, although the difference did not reach statistical significance.

4.1. Limitations

As professional players from one club were available for recruitment, a small sample resulted, and should be considered a limitation of this study. Although we ensured that participants completed no strenuous exercise or training in the 48 h before testing, we were unable to control for the amount of training that a player may have completed in the week leading into their testing session. This is one limitation of running research in a professional sporting setting. In high speed running and other fast movements there is an increased likelihood of an EMG sensor becoming dislodged and in this trial we missed data from the medial hamstrings of one participant as a result. This is a limitation of studying high speed running as opposed to a more controlled task.

Surface electromyography using telemetry allows for a non-invasive insight into muscle activity during overground activity. The measurements in this study were recorded using SENIAM guidelines from electrodes placed on a specific part of the muscles investigated. However, separate functional compartments within both of the hamstring muscles (Woodley & Mercer, 2005) and also gluteus maximus (Lyons, Perry, Gronley, Barnes, & Antonelli, 1983) have been identified. Consequently, the signal recorded may be biased towards one of these functional compartments and could potentially miss a change to the neuromuscular activity of the other compartments. It is also acknowledged that the use of surface EMG during high speed running in the field is unlikely to be as reliable as it might be at lower speeds of locomotion.

It has been suggested that hamstring muscle injuries are more likely to occur when an athlete is in a fatigued state (Woods et al., 2004). The testing procedure used in this study involved participants walking back between trials and was designed to keep players relatively fresh for each running effort. It is possible that due to the relatively fresh state of the athletes in this study that changes to the temporal characteristics of neuromuscular activation may have been harder to detect.

4.2. Future direction

The running in this study was confirmed using GPS to be 7.3 m s^{-1}.Typically any speed above 5.56 m s^{-1} is considered to be very high-speed running (Rampinini et al., 2015) and at speeds above 7 m s^{-1}, the strategy to increase running speed has been reported to shift from increasing stride length to increasing stride frequency, which requires increased muscle activation of the hip flexors, gluteus maximus and hamstring muscles (Dorn, Schache, & Pandy, 2012). Significantly, as running speed increases above 7 m s^{-1}, and approaches maximum speed, the gluteus maximus and hamstring muscles demonstrate the most dramatic increases in mechanical load and therefore increases in demands for muscular activation (Schache et al., 2011). Future research might consider assessing the temporal characteristics of muscular activation during overground running during maximum speed

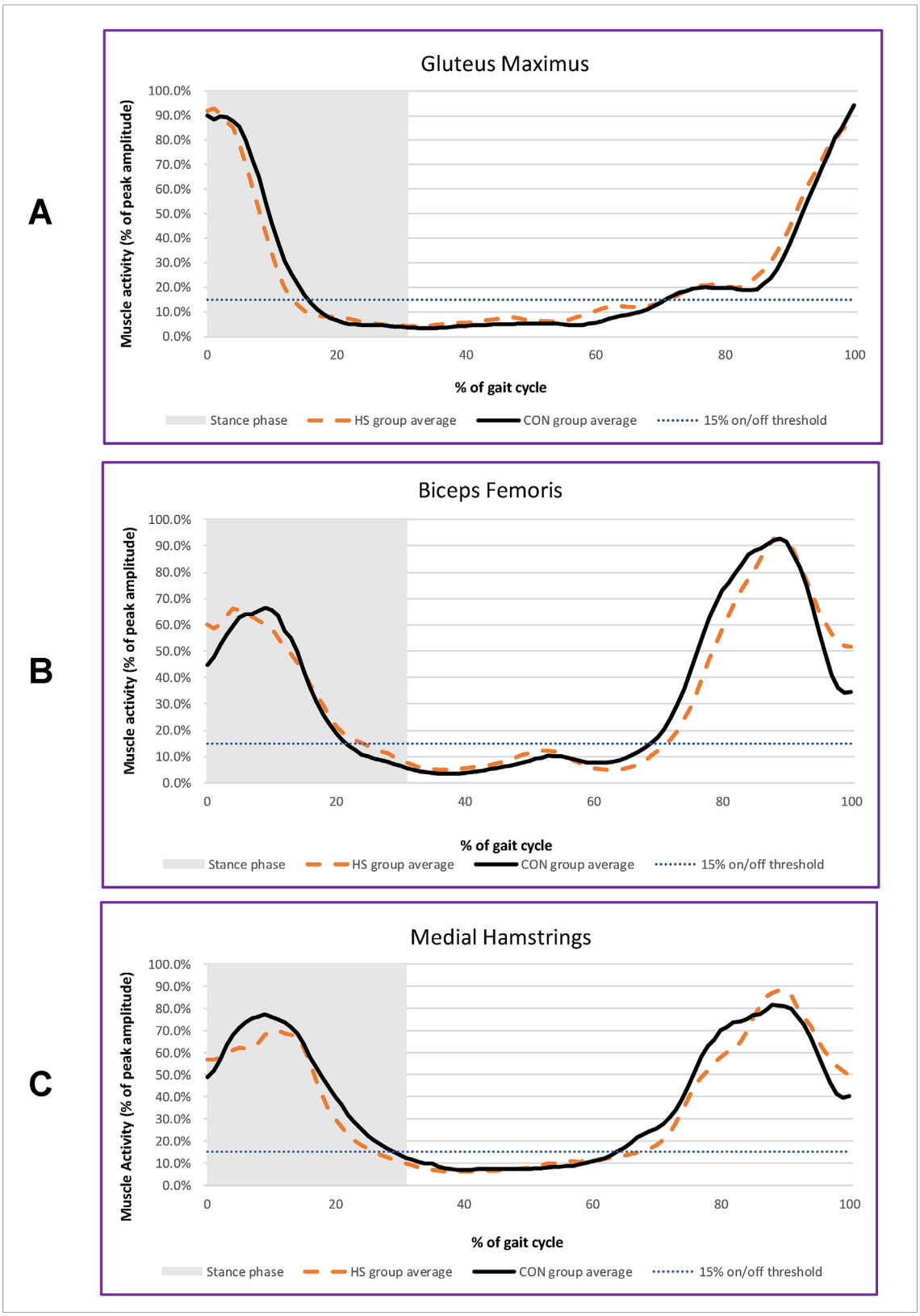

Fig. 1. Muscle activation pattern during high speed running: HS vs. CON group.
A = Gluteus Maximus muscle activation pattern; **B** = Biceps Femoris muscle activation pattern; **C** = Medial Hamstrings muscle activation pattern.

Table 2
Temporal characteristics of muscle activation during high speed running (% of gait cycle).

		HS Group Median (IQR)	CON group Median (IQR)	Effect Size	p
Gluteus maximus	onset	83.4 (5.2)	84.6 (6.4)	0.24	0.41
	offset	11.4 (2.7)	13.5 (2.8)	0.24	0.41
	duration	28.2 (4.6)	28.4 (3.7)	−0.03	0.95
Biceps Femoris	onset	70.0 (4.9)	68.8 (5.2)	−0.24	0.41
	offset	21.4 (8.8)	23.7 (8.6)	−0.07	0.85
	duration	52.8 (7.2)	56.5 (6.4)	0.24	0.41
Medial hamstrings	onset	71.0 (9.6)	64.4 (7.5)	−0.52	0.73
	offset	24.0 (9.1)	27.1 (9.1)	0.08	0.84
	duration	53.3 (8.9)	60.1 (11.2)	0.44	0.14

Abbreviation: IQR, Interquartile range.

running, or when athletes are in a more fatigued state, to determine if any neural inhibitory mechanisms are present when even greater demands are placed on the muscles involved. Given the results of this study we would not currently recommend assessment of muscle onset and offset times using EMG during overground running for the purpose of assessing re-injury risk following hamstring muscle injury. Practitioners may instead consider EMG assessment in protocols that have been reported to discriminate between injured and uninjured athletes such as isokinetic testing or a double leg to single leg stance task.

5. Conclusion

Following hamstring muscle injury no changes to the muscle onset time, offset time or duration of activity of the hamstring muscles or gluteus maximus were observed in professional Australian Football players during high-speed overground running.

Ethical approval

The study protocol was approved by the Faculty Human Ethics Committee (FHEC11/170), and all participants provided written informed consent.

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Declaration of competing interest

The authors declare no conflicts of interest. There was no grant aid, or manufacturers' aid, received to complete this study. The authors do not have any affiliations, or financial involvement, to declare with any company or organisation in the content or equipment discussed in the manuscript.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ptsp.2020.03.005.

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